```python
# Activity-1:

name = input("Enter your name : ")
num = input("Enter your contact number : ")
email = input("Enter your email id : ")
dob = input("Enter your birth date : ")

print(f"Welcome, {name}.")
print(f"Your contact number is : {num}")
print(f"{email} is your gmail id.")
print(f"You was born on {dob}.")

# Output:

# Enter your name : Cristiano Ronaldo
# Enter your contact number : 8843XX87XX
# Enter your email id : cristiano7@gmail.com
# Enter your birth date : 5 February, 1985.
# Welcome, Cristiano Ronaldo.
# Your contact number is : 8843XX87XX
# cristiano7@gmail.com is your gmail id.
# You was born on 5 February, 1985.
```

```python
# Activity-2:

# 2.1 Using print function.

print("Name = Maryam")
print("Contact = 7016340240")
print("Date of birth = 12 April,2009")

# 2.2 Using variable method.

name = "Maryam"
num = 7016340240
dob = "12 April,2009"

print("Name = ",name)
print("Contact = ",num)
print("Date of birth = ",dob)

# Output:

# Name = Maryam
# Contact = 7016340240
# Date of birth = 12 April,2009
```

```python
# Activity-3:

print("Maryam Bagia,\nB-7/1838 Ismail Manzil,\nSaiyedpura tunki,\nSurat.\nPin-395003")

# Output:

# Maryam Bagia,
# B-7/1838 Ismail Manzil,
# Saiyedpura tunki,
# Surat.
# Pin-395003
```

```python
# Activity-4:

print("""John said,"There's an elephant outside the window." """)


# Output:

# John said,"There's an elephant outside the window."
```

```
# Activity-5:

print("Mr.Patel,\nNear Avenue Complex,\nRing Road,\nSurat.\nPin-395003")

# Output:

# Mr.Patel,
# Near Avenue Complex,
# Ring Road,
# Surat.
# Pin-395003
```

```
# Activity-6:

# 6.1 Using print function directly.

print("Name = Maryam")
print("Course name = Computer Engineering")
print("Enrollment = 246150307008")

# 6.2 Using variable method.

name = "Maryam"
course = "Computer Engineering"
enroll = 246150307008

print("Name = ",name)
print("Course name = ",course)
print("Enrollment = ",enroll)

# Output:

# Name =  Maryam
# Course name =  Computer Engineering
# Enrollment =  246150307008
```

```
# Activity-7:

print("My name is Maryam.I am 16 years old.I am a student of Computer Science in Government Polytechnic for Girls,Sur

# Output:

# My name is Maryam.I am 16 years old.I am a student of Computer Science in Government Polytechnic for Girls,Surat.
# My current intrests are Android development and Software development.Mathematics and Programming are my favourite
# subject.My hobbies are cooking,coding and reading.I also like travelling and watching some educational or informati
# content on YouTube.I also like civix.If I get opportunity to prepare for any competitive exam in future, it will
# definately UPSC.Because, to become an IAS officer is my childhood dream. Thank you!
```

```python
# Activity-8:

print('"Life is 10% what happens to us and 90% how we react to it.\n- Charles R. Swindoll"')


# Output:

# "Life is 10% what happens to us and 90% how we react to it.
# - Charles R. Swindoll"
```

```python
# Activity-9:

print("Mr.Shah,\n\tNear Green Park,\n\tMG Road,\n\tMumbai\n\tPin-400001")

# Output:

# Mr.Shah,
#         Near Green Park,
#         MG Road,
#         Mumbai
#         Pin-400001
```

```python
# Activity-10:

quote = input("Enter your favourite quote : ")
writer = input("Whose quote is this? ")

print(f'Your Favourite Quote :\n-----------------\n"{quote}" - {writer}')


# Output:

# Your Favourite Quote :
# -----------------
# ""Talent without Working Hard is nothing."" - Cristiano Ronaldo
```

```python
# Activity-11:

print("Shopping List:\n1.Apples\n2.Bananas\n3.Grapes")

# Shopping List:
# 1.Apples
# 2.Bananas
# 3.Grapes
```

```python
# Activity-12:

print('Ramesh said,"Python is an easy-to-learn programming language."')

# Output:

# Ramesh said,"Python is an easy-to-learn programming language."
```

```python
# Activity-1:

i = 7
c = 24+8j
f = 7.01
m = [1,2,3]
t = (5,2,4)
k = {'a',7,9,'tree'}
s = "HELLO EVERYONE!!"
b = True
d = {
    'a':1,
    'b':2
}

print(i,type(i))
print(c,type(c))
print(f,type(f))
print(m,type(m))
print(t,type(t))
print(k,type(k))
print(s,type(s))
print(b,type(b))
print(d,type(d))

# Output:
# 7 <class 'int'>
# (24+8j) <class 'complex'>
# 7.01 <class 'float'>
# [1, 2, 3] <class 'list'>
# (5, 2, 4) <class 'tuple'>
# {'tree', 9, 'a', 7} <class 'set'>
# HELLO EVERYONE!! <class 'str'>
# True <class 'bool'>
# {'a': 1, 'b': 2} <class 'dict'>
```

```python
# Activity-4:

colors = {"Red","Blue","Purple","Turquoise","Black"}
print(colors)

# Output:
# {'Turquoise', 'Red', 'Black', 'Blue', 'Purple'}
```

```python
# Activity-5:

info = {
    "Maryam" : 99.21,
    "Mariyah" : 98.75,
    "Mubasshira" : 98.84
}
print(info)

# Output:
# {'Maryam': 99.21, 'Mariyah': 98.75, 'Mubasshira': 98.84}
```

```python
# Activity-2:

programming = ["Java","Python","Kotlin","JavaScript","SQL","C","C++","C#"]

print(programming)

# Output:

# ['Java', 'Python', 'Kotlin', 'JavaScript', 'SQL', 'C', 'C++', 'C#']
```

```python
# Activity-3:

countries = ("India","Russia","Saudi Arabia","Australia","UAE")
print(countries)

# Output:
# ('India', 'Russia', 'Saudi Arabia', 'Australia', 'UAE')
```

```python
# Activity-1:

a = float(input("Enter first number :"))
b = float(input("Enter second number :"))
c = float(input("Enter third number :"))

sum = a+b+c
avg = sum/3

print("Average of these three numbers = ",avg)

# Output:
# Enter first number :10
# Enter second number :20
# Enter third number :30
# Average of these three numbers =  20.0
```

```python
# Activity-2:

usd = float(input("Enter money in USD : "))

conversion_rate = 86.33
inr = usd*conversion_rate

print(f"{usd} USD is equals to {inr:.2f} INR.")

# Output:
# Enter money in USD : 100
# 100.0 USD is equals to 8633.00 INR.
```

```python
# Activity-3:

a = int(input("Enter value for a :"))
b = int(input("Enter value for b :"))

print(f"Before swapping : a = {a} , b = {b}.")

c = a
a = b
b = c

print(f"After swapping : a = {a} , b = {b}.")

# Output:
# Enter value for a :10
# Enter value for b :20
# Before swapping : a = 10 , b = 20.
# After swapping : a = 20 , b = 10.
```

```python
# Activity-4:

weight = float(input("Enter your weight in kg. : "))
height = float(input("Ente your height in meters : "))

BMI = weight/(height**2)

print(f"Your Body Mass Index (BMI) is {BMI:.2f} .")

# Output:
# Enter your weight in kg. : 70
# Enter your height in meters : 1.75
# Your Body Mass Index (BMI) is 22.86 .
```

```python
# Activity-5:

a = float(input("Enter first number :"))
b = float(input("Enter second number :"))
c = float(input("Enter third number :"))
d = float(input("Enter fourth number :"))
e = float(input("Enter fifth number :"))

sum = a+b+c+d+e
avg = sum/5

print(f"Sum = {sum} ,\nAverage = {avg}")

# Output:
# Enter first number :10
# Enter second number :20
# Enter third number :30
# Enter fourth number :40
# Enter fifth number :50
# Sum = 150.0 ,
# Average = 30.0
```

```python
# Activity-6:

cpf = int(input("Enter your CPF marks : "))
maths = int(input("Enter your Maths marks : "))
cse = int(input("Enter your CSE marks : "))
be = int(input("Enter your BE marks : "))
mp = int(input("Enter your MP marks : "))

sub_total = cpf + maths + cse + be + mp
total = 150
percentage = (sub_total/total)*100

print(f"Your percentage = {percentage:.2f}%")

# Output:
# Enter your CPF marks : 29
# Enter your Maths marks : 28
# Enter your CSE marks : 27
# Enter your BE marks : 29
# Enter your MP marks : 29
# Your percentage = 96.67%
```

```python
# Activity-1:

p = float(input("Enter Principal Amount : "))
r = float(input("Enter Rate of interest : "))
t = float(input("Enter Time period : "))

SI = (p*r*t)/100
CI = (p*((1+r/100))**t)-p

print(f"Simple Interest = {SI}")
print(f"Compound Interest = {CI}")

# Output:
# Enter Principal Amount : 1000
# Enter Rate of interest : 5
# Enter Time period : 2
# Simple Interest = 100.0
# Compound Interest = 102.5
```

```python
# Activity-2:

f = float(input("Enter temperature in Fahrenhit : "))

c = 5/9*(f - 32)

print(f"Temperature in celsius = {c:.2f}\u00B0C")

# Output:
# Enter temperature in Fahrenhit : 98.6
# Temperature in celsius = 37.00°C
```

```python
# Activity-4:

import math

radius = float(input("Enter the radius of the cylinder: "))
height = float(input("Enter the height of the cylinder: "))

volume = math.pi * (radius ** 2) * height

print(f"The volume of the cylinder = {volume:.2f}")

# Output:
# Enter the radius of the cylinder: 5
# Enter the height of the cylinder: 10
# The volume of the cylinder = 785.40
```

```python
# Activity-3:

import math

print("1.Circle")
print("2.Rectangle")
print("3.Triangle")

shape = input("Enter shape in lowercase :")

if(shape == "circle"):
    r = float(input("Enter radius of circle : "))
    area = math.pi*r**2
    print(f"Area of given circle = {area:.2f}")
elif(shape == "rectangle"):
    l = float(input("Enter length of rectangle : "))
    b = float(input("Enter breadth of rectangle : "))
    area = l*b
    print(f"Area of given rectangle = {area:.2f}")
else:
    base = float(input("Enter the base of the triangle: "))
    height = float(input("Enter the height of the triangle: "))
    area = 0.5 * base * height
    print(f"Area of the triangle = {area:.2f}")

# Output:
# Enter shape in lowercase :circle
# Enter radius of circle :5
# Area of given circle = 78.54
```

```python
# Activity-6:

x1 = float(input("Enter x1: "))
y1 = float(input("Enter y1: "))
x2 = float(input("Enter x2: "))
y2 = float(input("Enter y2: "))

slope = (y2 - y1) / (x2 - x1)

print(f"Slope of the line: {slope}")

# Output:
# Enter x1: 1
# Enter y1: 2
# Enter x2: 3
# Enter y2: 4
# Slope of the line: 1.0
```

```python
# Activity-5:

amount = float(input("Enter the amount: "))

rupees = int(amount)
paisa = int((amount - rupees) * 100)

print(f"Rupees: {rupees}, Paisa: {paisa}")

# Output:
# Enter the amount: 123.45
# Rupees: 123, Paisa: 45
```

```python
# Activity-1:

marks = float(input("Enter your marks (out of 100): "))

if marks >= 33:
    print("Congratulations! You have passed.")
else:
    print("Better luck next time but You have failed.")

# Output:
# Enter your marks (out of 100): 45
# Congratulations! You have passed.
```

```python
# Activity-2:

number = float(input("Enter a number: "))

if number > 0:
    print("The number is positive.")
elif number < 0:
    print("The number is negative.")
else:
    print("The number is zero.")

# Output:
# Enter a number: -5
# The number is negative.
```

```python
# Activity-3:

num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

if num1 >= num2 and num1 >= num3:
    largest = num1
elif num2 >= num1 and num2 >= num3:
    largest = num2
else:
    largest = num3

print(f"The largest number is: {largest}")

# Output:
# Enter the first number: 10
# Enter the second number: 20
# Enter the third number: 15
# The largest number is: 20.0
```

```python
# Activity-4:

year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")

# Output:
# Enter a year: 2020
# 2020 is a leap year.
```

```python
# Activity-5:

bill = float(input("Enter the purchase amount: "))

if(bill >= 5000):
    discount = bill * 0.20
elif(bill >= 3000):
    discount = bill * 0.10
else:
    discount = bill * 0.05

final_amount = bill - discount

print(f"Discount: {discount}")
print(f"Final amount to be paid: {final_amount}")

# Output:
# Enter the purchase amount: 6000
# Discount: 1200.0
# Final amount to be paid: 4800.0
```

```python
# Activity-6:

weight = float(input("Enter your weight in kg. : "))
height = float(input("Ente your height in meters : "))

BMI = weight/(height**2)

print(f"Your Body Mass Index (BMI) is {BMI:.2f} .")

if(BMI < 19):
    print("You are underweight.")
elif(19 < BMI < 25):
    print("You are normal weight.")
elif(BMI > 25):
    print("You are overweight.")

# Output:
# Enter your weight in kg. : 70
# Enter your height in meters : 1.75
# Your Body Mass Index (BMI) is 22.86.
# You are normal weight.
```

```python
# Activity-7:

marks = float(input("Enter your marks (out of 100): "))

if(marks>=90):
    print("You have scored an A grade.")
elif(80<=marks<89):
    print("You have scored a B grade.")
elif(70<=marks<79):
    print("You have scored a C grade.")
elif(60<=marks<69):
    print("You have scored a D grade.")
elif(marks<59):
    print("You have scored an E grade.")
else:
    print("You have scored an F grade.")

# Output:
# Enter your marks (out of 100): 85
# You have scored a B grade.
```

```python
# Activity-8:

import math

a = float(input("Enter coefficient a : "))
b = float(input("Enter coefficient b : "))
c = float(input("Enter coefficient c : "))

discriminant = b**2 - 4*a*c

if discriminant > 0:
    root1 = (-b + math.sqrt(discriminant)) / 2*a
    root2 = (-b - math.sqrt(discriminant)) / 2*a
    print(f"The roots are {root1} and {root2}")
elif discriminant == 0:
    root = -b / 2*a
    print(f"The root is {root}")
else:
    print("The equation has no real roots.")

# Output:
# Enter coefficient a : 1
# Enter coefficient b : -3
# Enter coefficient c : 2
# The roots are 2.0 and 1.0
```

```python
# Activity-1:

num = 1
while num <= 10:
    print(num)
    num += 1

# Output:
# 1
# 2
# 3
# 4
# 5
# 6
# 7
# 8
# 9
# 10
```

```python
# Activity-2

num = int(input("Enter a number: "))
factorial = 1

if num < 0:
    print("Factorial does not exist for negative numbers.")
elif num == 0:
    print("The factorial of 0 is 1.")
else:
    for i in range(1, num + 1):
        factorial *= i
    print(f"The factorial of {num} is {factorial}.")

# Output:
# Enter a number: 5
# The factorial of 5 is 120.
```

```python
# Activity-3:

number = input("Enter a number: ")

digit_sum = 0

i = 0
while i < len(number):
    digit_sum += int(number[i])
    i += 1

print("The sum of the digits is:", digit_sum)

# Output:
# Enter a number: 12345
# The sum of the digits is: 15
```

```python
# Activity-4:

number = int(input("Enter a number: "))

for i in range(1, 11):
    print(f"{number} X {i} = {number * i}")

# Output:
# Enter a number: 19
# 19 X 1 = 19
# 19 X 2 = 38
# 19 X 3 = 57
# 19 X 4 = 76
# 19 X 5 = 95
# 19 X 6 = 114
# 19 X 7 = 133
# 19 X 8 = 152
# 19 X 9 = 171
# 19 X 10 = 190
```

```python
# Activity-5:

for i in range(1,6):
    for j in range(1,i+1):
        print("*",end="")
    print()

# Output:
# *
# **
# ***
# ****
# *****
```

```python
# Activity-6:

sum=0
num=1

while num<=50:
    if num%2==0:
        sum+=1
    num+=1

print("The sum of even numbers from 1 to 50 is:",sum)

# Output:
#The sum of even numbers from 1 to 50 is: 25
```

```python
# Activity-7:

fact=0
n=int(input("enter a number:"))

for i in range(1,n+1):
    if n%i==0:
        fact+=1

if fact==2:
    print("%d is prime number"%(n))
else:
    print("%d is not prime number"%(n))

# Output:
# enter a number:47
# 47 is prime number
```

```python
# Activity-8:

for num in range(1, 101):
    if num%3==0 and num%5==0:
        print("%d"%(num))

# Output:
# 15
# 30
# 45
# 60
# 75
# 90
```

```python
# Activity-9:

reverse=0
n=int(input("enter a number:"))

while n>0:
    digit=n%10
    reverse= reverse*10+digit
    n//=10

print("reversed number:",reverse)

# Output:
# enter a number:12345
# reversed number: 54321
```

```python
# Activity-10:

#pattern-1                                  # Output:
for i in range(1,6):                        # *
    for j in range(1,i+1):                  # * *
        print("*",end=" ")                  # * * *
    print()                                 # * * * *
                                            # * * * * *


#pattern-2                                  # Output:
for i in range(1,6):                        # 1
    for j in range(1,i+1):                  # 1 2
        print(j,end=" ")                    # 1 2 3
    print()                                 # 1 2 3 4
                                            # 1 2 3 4 5


#pattern-3                                  # Output:
for i in range(1,6):                        # A
    j=65                                    # A B
    for j in range(j,j+i):                  # A B C
        print(chr(j),end=" ")               # A B C D
    print()                                 # A B C D E
```

```python
#pattern-4                                  # Output:
for i in range(5,0,-1):                     # * * * * *
    for j in range(1,i+1):                  # * * * *
        print("*",end=" ")                  # * * *
    print()                                 # * *
                                            # *


#pattern-5                                  # Output:
for i in range(5,0,-1):                     # 1 2 3 4 5
    for j in range(1,i+1):                  # 1 2 3 4
        print(j,end=" ")                    # 1 2 3
    print()                                 # 1 2
                                            # 1


#pattern-6                                  # Output:
for i in range(5,0,-1):                     # A B C D E
    j=65                                    # A B C D
    for j in range(j,j+i):                  # A B C
        print(chr(j),end=" ")               # A B
    print()                                 # A
```

```python
# Activity 1:

my_list = [10, 20, 30, 40, 50]

index1 = 1
index2 = 3

my_list[index1], my_list[index2] = my_list[index2], my_list[index1]
print("Swapped List:", my_list)

# Output:
# Swapped List: [10, 40, 30, 20, 50]
```

```python
# Activity-2:

my_list = [10, 20, 30, 40, 50]
total_sum = sum(my_list)

print("Sum of elements:", total_sum)

# Output:
# Sum of elements: 150
```

```python
# Activity-3:

my_list = [10, 20, 30, 40, 50]
element = 30

if element in my_list:
    print(f"{element} exists in the list.")
else:
    print(f"{element} does not exist in the list.")

# Output:
# 30 exists in the list.
```

```python
# Activity 4:

my_list = [12, 3, 5, 7, 19, 2, 8]
smallest = min(my_list)
largest = max(my_list)

print("Smallest element:", smallest)
print("Largest element:", largest)

# Output:
# Smallest element: 2
# Largest element: 19
```

```python
# Activity-5:

n=[1,2,3,4,5,6,7,8,9,10]

for number in n:
    if number %2==0:
        print("even no.=",number)

# Output:
# even no.= 2
# even no.= 4
# even no.= 6
# even no.= 8
# even no.= 10
```

```python
# Activity-6:

n=[1,4,9,16,25,36,49,64,81,100]

for number in n:
    if number%2!=0:
        print("odd no.=",number)

# Output:
# odd no.= 1
# odd no.= 9
# odd no.= 25
# odd no.= 49
# odd no.= 81
# odd no.= 100
```

```python
# Activity-7:

primes = [n for n in range(1, 51) if n > 1 and all(n % i != 0 for i in range(2, int(n**0.5) + 1))]
non_primes = [n for n in range(1, 51) if n not in primes]

print("Prime numbers:", primes)
print("Non-prime numbers:", non_primes)

# Output:
# Prime numbers: [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
# Non-prime numbers: [1, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36,
```

```python
# Activity-8:

lst = [10, 5, 8, 3, 7]

lst.append(12)
lst.remove(8)
print("List:", lst, "Length:", len(lst), "Index 2:", lst[2])

lst.sort()
print("Sorted:", lst)

lst.reverse()
print("Reversed:",lst)

# Output:
# List: [10, 5, 3, 7, 12] Length: 5 Index 2: 3
# Sorted: [3, 5, 7, 10, 12]
# Reversed: [12, 10, 7, 5, 3]
```

```python
# Activity-1:

my_set = {1, 2, 3}

# Add an element to the set
my_set.add(4)

# Remove an element from the set
my_set.remove(2)

# Get the length of the set
set_length = len(my_set)

print("Updated set:", my_set)
print("Length of the set:", set_length)

# Output:
# Updated set: {1, 3, 4}
# Length of the set: 3
```

```python
# Activity-2:

set1 = {1, 2, 3, 4, 5}
set2 = {4, 5, 6, 7, 8}

# Union
union_set = set1 | set2
print("Union:", union_set)

# Intersection
intersection_set = set1 & set2
print("Intersection:", intersection_set)

# Difference
difference_set = set1 - set2
print("Difference:", difference_set)

# Symmetric Difference
symmetric_difference_set = set1 ^ set2
print("Symmetric Difference:", symmetric_difference_set)

# Output:
# Union: {1, 2, 3, 4, 5, 6, 7, 8}
# Intersection: {4, 5}
# Difference: {1, 2, 3}
# Symmetric Difference: {1, 2, 3, 6, 7, 8}
```

```python
# Activity - 1

t=(10,20,30,'python',5.5,True)

print("first elememt:",t[0])
print("last element:",t[-1])
print("sliced tuple(index 1 to 3):",t[1:4])

# Output:
# first elememt: 10
# last element: True
# sliced tuple(index 1 to 3): (20, 30, 'python')
```

```python
# Activity - 2

t1=(1,2,3)
t2=('batch-1','batch-2','batch-3')

print("concatenation:",t1+t2)
print("repetation:",t2*2)

# Output:
# concatenation: (1, 2, 3, 'batch-1', 'batch-2', 'batch-3')
# repetation: ('batch-1', 'batch-2', 'batch-3', 'batch-1', 'batch-2', 'batch-3')
```

```python
# Activity - 3

my_tuple = (1, 2, 3, 4, 5)

element = int(input("Enter an element to check: "))

if element in my_tuple:
    print(f"{element} exists in the tuple.")
else:
    print(f"{element} does not exist in the tuple.")

if element not in my_tuple:
    print(f"{element} does not exist in the tuple.")
else:
    print(f"{element} exists in the tuple.")


# Output:
# Enter an element to check: 3
# 3 exists in the tuple.
```

```python
# Activity - 5

# Creating a tuple
my_tuple = (1, 2, 3, 4, 5)

# 1. Concatenating tuples
new_tuple = my_tuple + (6, 7)
print("Concatenated tuple:", new_tuple)

# 2. Repeating tuple
repeated_tuple = my_tuple * 2
print("Repeated tuple:", repeated_tuple)

# 3. Accessing elements
print("First element:", my_tuple[0])
print("Last element:", my_tuple[-1])

# 4. Slicing the tuple
print("Sliced tuple (2 to 4):", my_tuple[1:4])

# 5. Checking if an element exists
print("Is 3 in the tuple?", 3 in my_tuple)
print("Is 6 in the tuple?", 6 in my_tuple)

# 6. Finding the index of an element
print("Index of 4:", my_tuple.index(4))

# 7. Counting occurrences of an element
print("Count of 3:", my_tuple.count(3))

# 8. Tuple with mixed data types
mixed_tuple = (1, "Hello", 3.14, True)
print("Mixed tuple:",mixed_tuple)

# Output:
# Concatenated tuple: (1, 2, 3, 4, 5, 6, 7)
# Repeated tuple: (1, 2, 3, 4, 5, 1, 2, 3, 4, 5)
# First element: 1
# Last element: 5
# Sliced tuple (2 to 4): (2, 3, 4)
# Is 3 in the tuple? True
# Is 6 in the tuple? False
# Index of 4: 3
# Count of 3: 1
# Mixed tuple: (1, 'Hello', 3.14, True)
```

```python
# Activity - 4

my_tuple = (1, 2, 3, 4, 5, 3, 3)

# Length of the tuple
print("Length of the tuple:", len(my_tuple))

# Maximum value in the tuple
print("Maximum value in the tuple:", max(my_tuple))

# Count of an element in the tuple
element = 3
print(f"Count of {element} in the tuple:", my_tuple.count(element))

# Index of an element in the tuple
print(f"Index of {element} in the tuple:", my_tuple.index(element))

# Output:
# Length of the tuple: 7
# Maximum value in the tuple: 5
# Count of 3 in the tuple: 3
# Index of 3 in the tuple: 2
```

```python
# Activity - 1

string = input("Enter a string: ")
if string == string[::-1]:
    print("The string is a palindrome.")
else:
    print("The string is not a palindrome.")

# Output:
# Enter a string: madam
# The string is a palindrome.
```

```python
# Activity - 2

string = input("Enter a string: ")
vowels = "aeiouAEIOU"
count = 0
for char in string:
    if char in vowels:
        count += 1
print("Number of vowels:",count)

# Output:
# Enter a string: Hello World
# Number of vowels: 3
```

```python
# Activity - 3

string = input("Enter a string: ")
reversed_string = string[::-1]
print("Reversed string:", reversed_string)

# Output:
# Enter a string: Hello World
# Reversed string: dlroW olleH
```

```python
# Activity - 4

string1 = input("Enter first string: ")
string2 = input("Enter second string: ")

result = string1 + string2
print("Appended string:",result)

# Output:
# Enter first string: Hello
# Enter second string: World
# Appended string: HelloWorld
```

```python
# Activity - 5

string = input("Enter a string: ")
char = input("Enter a character to count: ")

count = string.count(char)
print("Occurrence of", char,":",count)

# Output:
# Enter a string: Hello World
# Enter a character to count: o
# Occurrence of o : 2
```

```python
# Activity - 6

string = input("Enter a string: ")
string_without_spaces = string.replace(" ", "")
print("String without spaces:", string_without_spaces)

# Output:
# Enter a string: Hello World
# String without spaces: HelloWorld
```

```python
# Activity - 1

dict1 = {'a': 1, 'b': 2}
dict2 = {'c': 3, 'd': 4}

dict1.update(dict2)
print(dict1)

# Output:
# {'a': 1, 'b': 2, 'c': 3, 'd': 4}
```

```python
# Activity - 2

temperature = {'Monday': 22, 'Tuesday': 28, 'Wednesday': 19, 'Thursday': 24, 'Friday': 30}

filtered_temp = {day: temp for day, temp in temperature.items() if 20 <= temp <= 25}
print(filtered_temp)

# Output:
# {'Monday': 22, 'Thursday': 24}
```

```python
# Activity - 3

dict1 = {'a': 1, 'b': 2, 'c': 3}

# Adding an item
dict1['d'] = 4

# Updating an item
dict1['b'] = 5

# Removing an item
dict1.pop('a')

# Getting a value for a key
value = dict1.get('c')

# Checking if a key exists
key_exists = 'd' in dict1

# Clearing all items
dict1.clear()

print("Updated Dictionary:", dict1)
print("Value of 'c':", value)
print("Does 'd' exist?",key_exists)

# Output:
# Updated Dictionary: {}
# Value of 'c': 3
# Does 'd' exist? True
```

```python
# Activity - 1

def factorial(n):
    result = 1
    for i in range(1, n+1):
        result *= i
    return result

num = int(input("Enter a number: "))
print("Factorial of", num, "is",factorial(num))

# Output:
# Enter a number: 5
# Factorial of 5 is 120
```

```python
# Activity - 2

def check_even_odd(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"

num = int(input("Enter a number: "))
print(num, "is", check_even_odd(num))

# Output:
# Enter a number: 5
# 5 is Odd
```

```python
# Activity - 3

def area_of_circle(radius):
    return 3.14 * radius * radius

radius = float(input("Enter the radius of the circle: "))
print("Area of the circle is", area_of_circle(radius))

# Output:
# Enter the radius of the circle: 5
# Area of the circle is 78.5
```

```python
# Activity - 4

def largest_number(a, b, c):
    return max(a, b, c)

num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

print("The largest number is", largest_number(num1,num2,num3))

# Output:
# Enter the first number: 5
# Enter the second number: 10
# Enter the third number: 7
# The largest number is 10.0
```

```python
# Activity - 6

def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n-1) + fibonacci(n-2)

terms = int(input("Enter the number of terms: "))

print("Fibonacci sequence:")
for i in range(terms):
    print(fibonacci(i),end=" ")

# Output:
# Enter the number of terms: 10
# Fibonacci sequence:
# 0 1 1 2 3 5 8 13 21 34
```

```python
# Activity - 9

def sort_string(s):
    return ''.join(sorted(s))

string = input("Enter a string: ")
print("Sorted string:", sort_string(

# Output:
# Enter a string: python
# Sorted string: hnopty
```

```python
# Activity - 7

def gcd(a, b):
    if b == 0:
        return a
    else:
        return gcd(b, a % b)

num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))

print("GCD of", num1, "and", num2, "is", gcd(num1,num2))

# Output:
# Enter the first number: 56
# Enter the second number: 98
# GCD of 56 and 98 is 14
```

```python
# Activity - 8

def decimal_to_hexadecimal(decimal):
    return hex(decimal)

num = int(input("Enter a decimal number: "))
print("The hexadecimal representation is", decimal_to_hexadecimal(num))

# Output:
# Enter a decimal number: 255
# The hexadecimal representation is 0xff
```

```python
# Activity - 1

import random
import math
import datetime

# Using random module to generate a random number
random_number = random.randint(1, 100)
print("Random number between 1 and 100:", random_number)

# Using math module to calculate square root
num = 16
sqrt_num = math.sqrt(num)
print(f"Square root of {num} is {sqrt_num}")

# Using datetime module to get the current date and time
current_datetime = datetime.datetime.now()
print("Current date and time:", current_datetime)

# Output:
# Random number between 1 and 100: 42
# Square root of 16 is 4.0
# Current date and time: 2025-05-29 12:34:56.789012
```

custommodule.py

Practical 14 > custommodule.py > ...

```python
# Activity - 2
# custommodule.py

def reverse_string(s):
    return s[::-1]

def capitalize_string(s):
    return s.capitalize()

def string_length(s):
    return len(s)

```

Question 2.py

Practical 14 > Question 2.py > ...

```python
# Activity - 2
# main.py

import custommodule

string = input("Enter a string: ")

reversed_string = custommodule.reverse_string(string)
capitalized_string = custommodule.capitalize_string(st
length_of_string = custommodule.string_length(string)

print("Reversed string:", reversed_string)
print("Capitalized string:", capitalized_string)
print("Length of string:", length_of_string)

# Output:
# Enter a string: hello world
# Reversed string: dlrow olleh
# Capitalized string: Hello world
# Length of string: 11

```

```python
# Activity - 3

import math

def compute_mean(numbers):
    return sum(numbers) / len(numbers)

def compute_standard_deviation(numbers, mean):
    variance = sum((x - mean) ** 2 for x in numbers) / len(numbers)
    return math.sqrt(variance)

numbers = list(map(int, input("Enter numbers separated by space: ").split()))

mean = compute_mean(numbers)
std_dev = compute_standard_deviation(numbers, mean)

print("Mean:", mean)
print("Standard Deviation:",std_dev)

# Output:
# Enter numbers separated by space: 1 2 3 4 5
# Mean: 3.0
# Standard Deviation: 1.4142135623730951
```

```python
# Activity - 4

import random

def rock_paper_scissors(player_choice):
    choices = ['rock', 'paper', 'scissors']
    computer_choice = random.choice(choices)

    if player_choice == computer_choice:
        return f"It's a tie! Both chose {player_choice}."

    if (player_choice == 'rock' and computer_choice == 'scissors') or \
       (player_choice == 'paper' and computer_choice == 'rock') or \
       (player_choice == 'scissors' and computer_choice == 'paper'):
        return f"You win! {player_choice} beats {computer_choice}."

    return f"You lose! {computer_choice} beats {player_choice}."

player_choice = input("Enter your choice (rock, paper, or scissors): ").lower()
while player_choice not in ['rock', 'paper', 'scissors']:
    player_choice = input("Invalid choice. Please enter rock, paper, or scissors: ").lower()

result = rock_paper_scissors(player_choice)
print(result)

# Output:
# Enter your choice (rock, paper, or scissors): rock
# You win! rock beats scissors.
```

```python
# Activity - 1

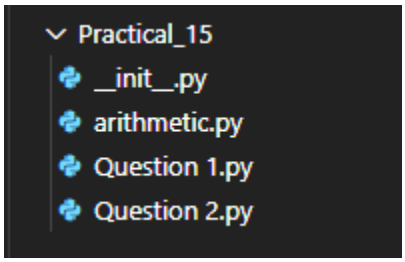
import math

# Square root
num = float(input("Enter a number to find its square root: "))
square_root = math.sqrt(num)
print(f"Square root of {num} is {square_root}")

# Power
base = float(input("Enter the base number: "))
exponent = float(input("Enter the exponent: "))
power = math.pow(base, exponent)
print(f"{base} raised to the power of {exponent} is {power}")

# Logarithm
log_num = float(input("Enter a number to find its logarithm (base 10): "))
log_value = math.log10(log_num)
print(f"Logarithm (base 10) of {log_num} is {log_value}")

# Output:
# Enter a number to find its square root: 16
# Square root of 16 is 4.0
# Enter the base number: 2
# Enter the exponent: 3
# 2 raised to the power of 3 is 8.0
# Enter a number to find its logarithm (base 10): 100
# Logarithm (base 10) of 100 is 2.0
```



arithmetic.py

Practical_15 > arithmetic.py > ...

```python
# Activity - 2

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Error! Division by zero."

```

Question 2.py

Practical_15 > Question 2.py > ...

```python
# Activity - 2

from Practical_15 import arithmetic

a = float(input("Enter the first number: "))
b = float(input("Enter the second number: "))

print(f"The sum of {a} and {b} is: {arithmetic.add(a, b)}")
print(f"The difference of {a} and {b} is: {arithmetic.subtract(a, b)}")
print(f"The product of {a} and {b} is: {arithmetic.multiply(a, b)}")
print(f"The division of {a} by {b} is: {arithmetic.divide(a,b)}")

# Output:
# Enter the first number: 10
# Enter the second number: 5
# The sum of 10.0 and 5.0 is: 15.0
# The difference of 10.0 and 5.0 is: 5.0
# The product of 10.0 and 5.0 is: 50.0
# The division of 10.0 by 5.0 is: 2.0
```

```python
# Activity - 1

paragraph = input("Enter a paragraph: ")

with open("paragraph.txt", "w") as file:
    file.write(paragraph)

with open("paragraph.txt", "r") as file:
    content = file.read()

print("Content of the file:")
print(content)
```

```python
# Activity - 2

text_to_append = input("Enter text to append to the file: ")

with open("paragraph.txt", "a") as file:
    file.write("\n" + text_to_append)

with open("paragraph.txt", "r") as file:
    content = file.read()

print("Updated content of the file:")
print(content)
```

```python
# Activity - 3

with open("paragraph.txt", "r") as file:
    content = file.read()

word_count = len(content.split())

print("Number of words in the file:",word_count)
```

```python
# Activity - 4

with open("paragraph.txt", "r") as file:
    for line in file:
        print(line,end="")
```

```python
# Activity - 5

with open("source.txt", "r") as source_file:
    content = source_file.read()

with open("destination.txt", "w") as destination_file:
    destination_file.write(content)
```